# Topology of Neural-Network Layers for Out-of-Distribution Detection

Dat Bui · October 2026 · code and full results: github.com/Datquangbui1011/Research


## At a glance

**Task:** find a model used in the six papers of the literature report, implement all six approaches on it, use them to detect out-of-distribution (OOD) inputs, and compare how well each works.

**Model:** the Bayesian MLP 784→64→16→10 from the pTU paper [3], trained on MNIST (97.2 % test accuracy, close to the paper's 97.4 %).

**Test:** can each method tell MNIST digits apart from 7 kinds of unfamiliar images? Every result is averaged over 3 independently trained models.

**Main result:** 5 of the 6 topological approaches beat the standard confidence baseline. The best one, [5], reaches 96.5 / 100 AUROC versus 74.1 for confidence. Neural Persistence [1] cannot detect OOD inputs at all.


## 1. The shared model

- **Why this model:** papers [1], [2] and [3] all study small fully connected networks (MLPs) on MNIST, and [3] gives the exact architecture, prior and training rule, so the setup can be reproduced.
- **What it is:** 784 inputs (28×28 pixels) → 64 → 16 → 10 classes, ReLU, trained as a Bayesian network (Bayes-by-Backprop), so each weight has a distribution, not a single value.
- **How it is used:** methods [1], [2], [4], [5], [6] use the average weights; [3] (pTU) uses 30 random weight samples, as in the paper.
- **One change from the paper:** the KL term of the loss is scaled by 0.1. With the full term, accuracy stalls at 94.5 %, and [3] does not say how it weighted this term.

## 2. The six approaches

All six turn a layer of the network into a shape (a graph or a cloud of points), track how that shape connects up as a threshold changes (persistent homology), and summarize it with numbers. An input whose numbers look unusual compared with training data is flagged as OOD.

| Paper | Main idea in one sentence | OOD score used here |
|---|---|---|
| [1] Neural Persistence<br>Rieck et al. 2019 | How strongly a few weights stand out in each layer of the trained network (uses weights only, no input). | Same number computed on each input's activations (my adaptation, since the original ignores the input). |
| [2] Topological Uncertainty<br>Lacombe et al. 2021 | Each input activates the network in a pattern; compare that pattern's topology with typical training inputs. | Distance to the average training pattern of the predicted class. |
| [3] pTU<br>Yeh & Yang 2025 | Run the input through 30 sampled versions of a Bayesian network; if their topologies disagree, the input is uncertain. | Spread (variance) of the 30 topologies; plus the paper's statistical test on groups of images. |
| [4] Persistent Laplacian<br>Wolf et al. 2025 | Eigenvalues of a matrix built from the shape: zero eigenvalues = topology, nonzero ones = extra geometry. | Number of zero eigenvalues ($\beta_0$) and the nonzero eigenvalues at 3 thresholds. |
| [5] Data manifold<br>Magai & Ayzenberg 2022 | Treat all images at one layer as a point cloud and measure its loops, clusters and dimension. | Groups of images: cloud topology + dimension. Single image (my adaptation): how much it changes its 20 nearest training neighbours' topology. |
| [6] CNN activations<br>Purvine et al. 2023 | Compare the topology of different layers' point clouds with a distance between diagrams. | Groups of images: distance from the group's diagrams to MNIST's diagrams. |

## 3. How the methods were tested

- **Normal data:** 2000 MNIST test digits. **OOD data** (2000 each, all shown as 28×28 grayscale): Fashion-MNIST (clothes), KMNIST (Japanese characters), CIFAR-10 (photos), SVHN (house numbers), Gaussian noise, uniform noise, and MNIST-flip (digits with 100 pixels inverted).
- **Score:** AUROC = how well a method ranks OOD images above normal ones. **50 = random guessing, 100 = perfect.** FPR95 (lower is better) = % of OOD images missed when 95 % of normal images are accepted.
- **Two settings: one image at a time** (the usual OOD task) and **groups of images** (is this whole batch from a different source?), which is how papers [3], [5], [6] use topology.
- **Baselines:** softmax confidence (MSP, the baseline in [2]), Bayesian mutual information, and distance to the nearest training images (k-NN).

## 4. Results

**4a. One image at a time** — average over the 7 OOD sets, mean ± std over 3 trained models, ranked from best to worst. Grey rows are baselines.

| # | Method | AUROC ↑ | FPR95 ↓ | Verdict |
|---|---|---|---|---|
| 1 | [5] Data manifold, single-image version | **96.5 ± 0.4** | **16.9 ± 2.3** | **Best** — but it mainly measures distance to the training data |
| 2 | k-NN distance (baseline) | 88.4 ± 0.9 | 46.4 ± 1.9 | **Strong baseline** |
| 3 | [2] TU with layers rescaled (my fix) | 87.4 ± 0.6 | 60.1 ± 1.6 | **Good** — +8 points over the paper's version |
| 4 | [4] Persistent Laplacian, zero eigenvalues only ($\beta_0$) | 85.8 ± 0.1 | 50.3 ± 1.6 | **Good** — very stable |
| 5 | [3] pTU | 83.0 ± 5.2 | 39.4 ± 7.0 | **Good but unstable** — misses Gaussian noise |
| 6 | [4] Persistent Laplacian, full spectrum | 82.6 ± 0.8 | 59.3 ± 1.5 | **OK** — extra eigenvalues make it worse |
| 7 | Bayesian mutual information (baseline) | 81.4 ± 4.8 | 60.2 ± 9.0 | **Baseline** |
| 8 | [2] TU (paper version) | 79.1 ± 0.8 | 73.1 ± 3.1 | **OK** — beats confidence |
| 9 | Softmax confidence, MSP (baseline) | 74.1 ± 4.8 | 74.1 ± 3.0 | **Weak baseline** — fooled by confident mistakes |
| 10 | [1] Neural Persistence | 55.4 ± 5.0 | 89.2 ± 4.9 | **Fails** — near random guessing |

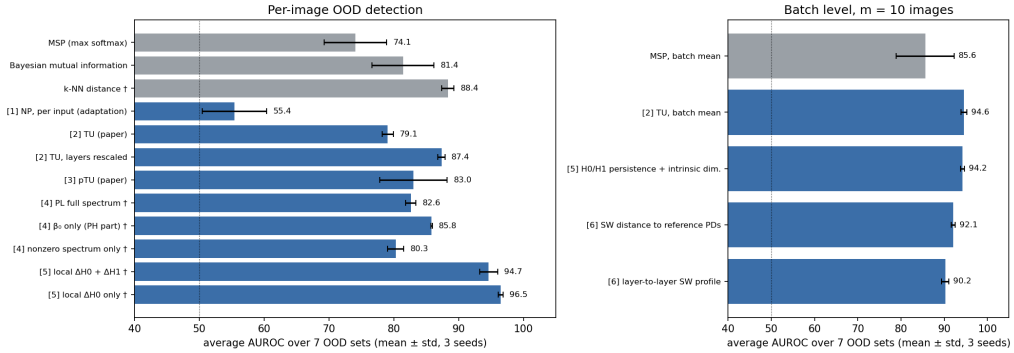



Figure 1. Average AUROC (mean ± std over 3 models). Grey = baselines, blue = topological approaches; † = evaluated on 1000 images per set. Dotted line = random guessing.

**4b. Groups of images** — is a whole batch of m images from a different source? (average AUROC, 3 models)

| Method | 10 images | 20 images | Comment |
|---|---|---|---|
| [2] TU, averaged over the batch | **94.6 ± 0.7** | 96.8 ± 0.5 | Best with very small batches |
| [5] Cloud topology + intrinsic dimension | 94.2 ± 0.5 | 98.9 ± 0.0 | Ties TU at 10 images, ahead from 20 |
| [6] Distance between persistence diagrams | 92.1 ± 0.4 | **99.0 ± 0.2** | Near perfect from 20 images |
| [6] Layer-to-layer distance profile | 90.2 ± 0.9 | 94.6 ± 1.6 | Paper's main tool; weaker for OOD |
| Softmax confidence, averaged (baseline) | 85.6 ± 6.7 | 89.0 ± 8.3 | Lowest and least stable |

**[3] pTU statistical test** (the paper's own test on groups of images): with 50 images it flags every OOD set in 96–100 % of trials, while flagging normal MNIST only 3–5 % of the time (the intended 5 % false-alarm rate). All group methods reach about 100 AUROC with 50 images.

## 5. Main takeaways

1. **Topology catches what confidence misses.** The network is 94–100 % confident on SVHN, CIFAR-10 and uniform noise, so the confidence score fails there (AUROC 60–69). TU, pTU and $\beta_0$ still detect them (78–100), and their results barely change between trained models.
2. **A method must look at the input.** Neural Persistence [1] describes the trained weights, which are the same for every input, so it cannot tell normal and unusual inputs apart.
3. **Simple summaries beat complicated ones.** Counting connected pieces ($\beta_0$) works better than the full persistent-Laplacian spectrum and TU's distance; the extra eigenvalues and loop (H1) features add noise.
4. **How layers are combined matters.** The first layer is the most informative in [2], [3] and [4], but those papers average all layers equally. Rescaling each layer before averaging lifts TU from 79 to 87.
5. **The best single-image score is mostly a distance.** [5]'s top result measures how far an image is from its nearest training images; its loop-based part alone is worse than guessing.
6. **For groups of images, everything works.** All group methods reach about 100 AUROC with 50 images; pTU's test is the only one with a controlled false-alarm rate.

## 6. Limitations and next steps

- **Scope:** one small MLP on MNIST (the setting of [1]–[3]; [5] and [6] studied CNNs); 3 trained models.
- **My adaptations** (labelled above): single-image versions of [1] and [5], the MLP version of [6], rescaled TU, and $\beta_0$-only; plus the 0.1 KL scaling.
- **Next:** repeat on a CNN (CIFAR-10), test layer-weighted TU/pTU, and use a two-sided pTU score so low-variance inputs such as Gaussian noise are caught.

References: [1] Rieck et al., ICLR 2019. [2] Lacombe et al., IJCAI 2021. [3] Yeh & Yang, arXiv:2511.18813. [4] Wolf et al., arXiv:2509.20220; Jones & Wei (PETLS); Botnan & Dong, arXiv:2408.16741. [5] Magai & Ayzenberg, arXiv:2204.08624. [6] Purvine et al., AAAI 2023.